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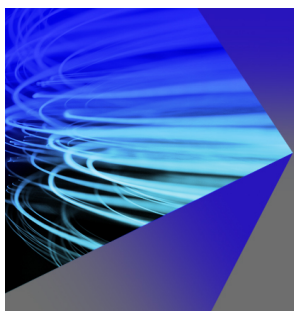
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ABSTRACT

The electromagnetic design optimization of a hybrid central solenoid (CS) coil is studied for a medium-sized superconducting fusion tokamak, which is smaller than the China Fusion Engineering Test Reactor and aims to realize a fusion power gain $Q > 1$. The CS coil uses ReBCO and Nb₃Sn conductors in the high and low field sub-coils, which are both composed of six vertically stacked modules with unequal heights. Based on the medium-sized tokamak device, a code is written to maximize the magnetic flux, meanwhile constraining the peak fields, WP height, and the total conductor length. An optimization design of the hybrid CS coil is obtained using the code. Then, the electromagnetic field at initial energization ($t = 0$ s) and fast discharge is calculated using Maxwell and MATLAB, respectively. The operating current exhibits exponential decay to zero during fast discharge, with the current decrease rates of the high-temperature superconductivity and low-temperature superconductivity sub-coils determined by their inductances. The Maxwell electromagnetic analysis shows good agreement with the results of the self-written code. In addition, a 2D axisymmetric finite element model, incorporating cables, conductor jackets, and insulation, is developed and simulated using ANSYS's coupled structural–thermal–electromagnetic solver.

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I. INTRODUCTION

The central solenoid (CS) coil is an essential component in next-generation fusion tokamaks, tasked with providing the magnetic flux necessary to initiate and sustain plasma.^{6,7} This paper presents an electromagnetic design study of a hybrid central solenoid for a medium-sized superconducting tokamak, which is strategically positioned for construction before the first plasma operation of the International Thermonuclear Experimental Reactor (ITER) and the development of the China Fusion Engineering Test Reactor (CFETR).^{1–5} This medium-sized tokamak features a major radius of 3.5 m, a minor radius of 1.1 m, and a toroidal magnetic field (B_t) of 5.3 T, with the aim to achieve a fusion power gain (Q) greater than 1.

The structure of this paper is as follows: In Sec. II, we introduce the design of the CS conductors. Subsequently, to enhance fusion performance by maximizing magnetic flux, we explore the optimization of the CS coil design in Sec. III.^{8,9} A MATLAB-based code has been written to maximize the magnetic flux, while

adhering to specified constraints. Then, in Sec. IV, the electromagnetic analysis is performed, and the current and vertical force on each sub-coil during fast discharge are calculated. In Sec. V, a preliminary mechanical analysis of the conductor jacket is conducted. Finally, Sec. VI provides conclusions.

II. THE HTS AND LTS CONDUCTORS

Based on the conductor design in the Institute of Plasma Physics, Chinese Academy of Sciences (ASIPP), ReBCO^{10,11} and high- J_c Nb₃Sn¹² conductors are used for the high field (HF) and low field (LF) sub-coils, respectively, as shown in Fig. 1.

The circle-in-square high-temperature superconductivity (HTS) conductor consists of six embedded sub-cables. The cable is enclosed within a stainless steel jacket. Within the 31.2 mm diameter cable region, six sub-cables are evenly distributed, and each sub-cable is wrapped with 32 ReBCO tapes and eight copper tapes.¹³ To take away the heat generated by AC operation of the CS coil, a copper die is used to fabricate

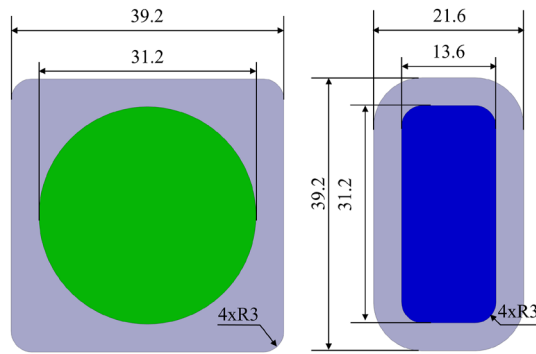


FIG. 1. ReBCO and high- J_c Nb_3Sn conductors.

the ReBCO cable,¹⁴ and the void fraction of the ReBCO cable is 29%.

The rectangular LTS conductor has a size of $31.2 \times 13.6 \text{ mm}^2$ and consists of 480 high- J_c Nb_3Sn strands, each with a diameter of 0.82 mm.^{15–17} The void fraction of the low-temperature superconductivity (LTS) cable is 27%.

Based on the conductor design, considering the temperature margin, the allowable peak field for the HTS conductor is 18.5 T at an operational current of 39.5 kA, which is 12.4 T for the LTS conductor.¹⁸ The detailed parameters of the conductors¹⁹ are listed in Table I.

The CS conductors will experience a huge Lorentz force generated by the strong magnetic field,²⁰ which needs to be withstood by the jacket. Hence, the high strength Nitronic 50 stainless steel (N50H) developed by ASIPP is chosen as the jacket material.^{21,22} In our preliminary design and analysis stage, the thicknesses of the HTS and LTS conductor jackets have been iteratively optimized to 4 mm. To reduce variables during the optimization process, we fixed the jacket thickness at 4 mm. In addition, subsequent mechanical analysis, performed after the electromagnetic optimization, further confirmed that the 4 mm thickness was suitable.

TABLE I. Detailed parameters of the conductors.

Parameters	HTS	LTS
Superconductor	ReBCO	High- J_c Nb_3Sn
Cable size (mm)	31.2	31.2×13.6
Diameter of sub-cable (mm)	9	...
Number of sub-cables	6	...
Number of tapes in sub-cable	32	...
Tape size (mm)	4×0.1	...
Strand diameter (mm)	...	0.82
Number of SC strands	...	480
Ratio of copper to SC	...	1:1
Jacket thickness (mm)	4	4
Void fraction	29%	27%

III. OPTIMIZATION OF THE ELECTROMAGNETIC DESIGN

A cross-sectional 1/2 CS model is established for the design study, as shown in Fig. 2. The radial build of the next generation tokamak allocates the CS coil winding packs in a radius of 1 m. Because the optimization objective of the CS coil design is to maximize the magnetic flux, the CS coil uses HTS conductors in the high field region to enhance the magnetic field and, hence, to increase the magnetic flux. The HTS and the LTS sub-coils are both composed of six stacked modules with unequal heights,²³ half of which are labeled 3U, 2U, and 1U. Because the 1U modules are more effective at producing magnetic flux, the HTS1U and LTS1U modules are designed to comprise more axial turns than the 2U and 3U modules.

A 25 mm radial gap is intentionally left between the HTS and LTS sub-coils, which will remain unfilled. The HTS and LTS coils will be preloaded independently, with each coil bearing its own radial electromagnetic forces without transmitting these forces between them.²⁴

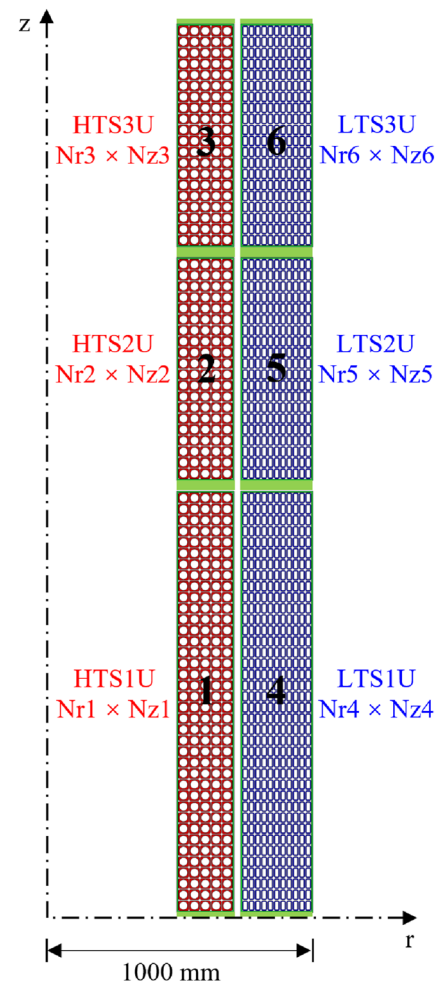


FIG. 2. Schematic of the CS coil optimization model.

The number of turns of conductors in the radial and axial directions is represented as N_r and N_z . The modules of the HTS (or the LTS) sub-coil have the same N_r .²⁵ The radially adjacent HTS and LTS modules have the same N_z .

In addition, the 3U and 2U modules of the HTS (or the LTS) sub-coil are wound with the same number of radial and axial turns of conductors ($N_{r1} = N_{r2} = N_{r3}$, $N_{r4} = N_{r5} = N_{r6}$, $N_{z1} = N_{z4}$, and $N_{z2} = N_{z3} = N_{z5} = N_{z6}$).

The HTS and LTS sub-coils are insulated as follows: 1 mm turn insulation, 0.5 mm layer insulation, 1 mm pancake insulation, 6 mm ground insulation, and a 40 mm G10 spacer placed between the adjacent modules.

A MATLAB code is written to perform the electromagnetic design optimization of the CS coil, and the flow chart is shown in Fig. 3. First, the coil parameters are initialized, such as the outer radius, radial gap, cable size, jacket thickness, insulation thickness, and operation current. Next, all the candidate designs ($6 \times 10 \times 21 \times 21 = 26\,460$) are modeled and calculated in turn by varying the numbers of turns (N_{r1} , N_{r4} , N_{z1} , and N_{z2}) through the optimization range.

Then, the constraints including the peak field, the coil height, and the conductor length are evaluated, as shown in Fig. 3.

In our code, each cable of the CS coil is modeled with 81 circular current filaments; the field contribution of each current filament is accumulated to obtain the magnetic field on the CS coil by the Biot-Savart law.²⁶ For the peak fields, considering the structure and symmetry of the CS coil,²⁷ the fields on the cables of the innermost layer of HTS1U or LTS1U are calculated, and the maximum one is picked out as the peak field. Considering the operation temperature margin for the conductors, the allowable peak fields²⁸ for the HTS and LTS conductors are 18.5 and 12.4 T, respectively.^{29,30}

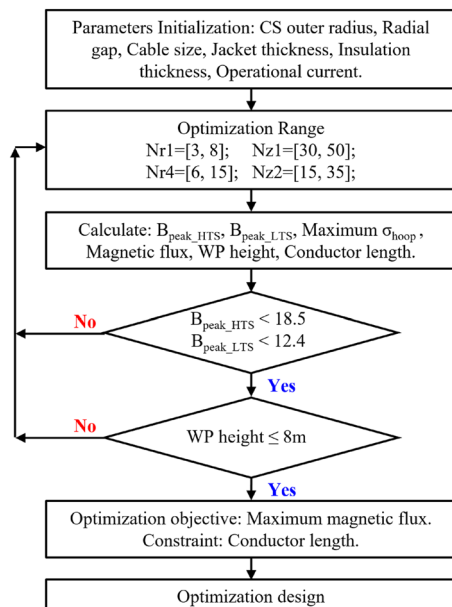


FIG. 3. Flow chart of the MATLAB code for the electromagnetic design optimization of the CS coil.

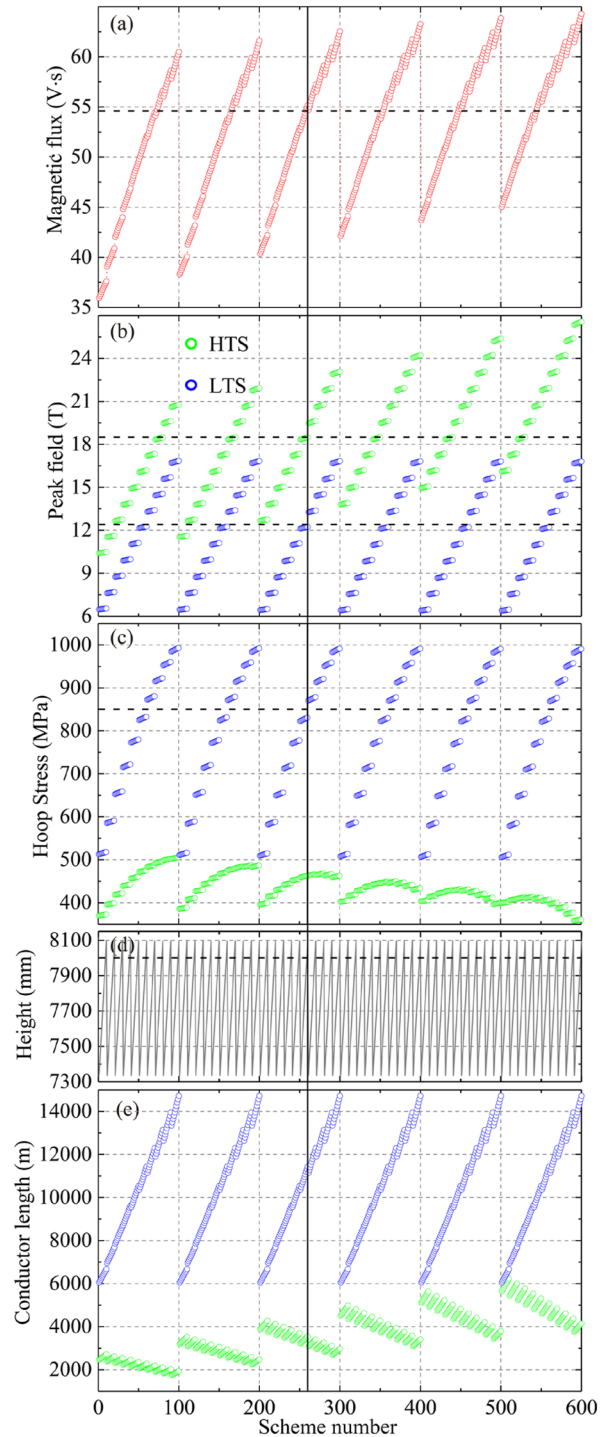


FIG. 4. (a) Magnetic flux, (b) peak field, (c) hoop stress, (d) WP height, and (e) conductor length illustration of the CS coil design optimization. Only 600 out of 26 460 candidate designs are shown. The black dashed lines represent the allowable peak field, hoop stress, and WP height of the CS coil. The vertical black line represents the final optimization design. The fields (HTS and LTS conductors) and WP height have all reached the maximum allowable limits.

For the height^{31,32} of CS coil WP, based on the relationship between the heights of the CS and TF coils,³³ the allowable maximum height is set at 8 m.³⁴

Finally, among the designs that satisfy the above constraints, the one with the highest magnetic flux is selected as the optimization design for the CS coil. For the designs that produce comparable magnetic flux, the one with the shorter total conductor length is chosen to minimize cost.

To facilitate the display of results, only the results corresponding to 600 designs are presented by limiting the range of N_{r1} , N_{r4} , and N_{z1} , with graphs plotted accordingly, as shown in Fig. 4. From the designs displayed, it is evident that when the peak field increases, the magnetic flux increases correspondingly. In Figs. 4(b) and 4(d), the black dashed lines represent the requirements for the peak field of the HTS and LTS conductors and the height of CS coil WP, respectively. The final optimization design is No. 260, which is indicated by the vertical black line through Fig. 4, its peak.

The 3D map shows the calculation results of 26 460 designs in Fig. 5. Each sphere represents a candidate design, the color represents the level of the magnetic flux, and the size of the spheres represents the total conductor length of the CS coil. Meanwhile, the figure delineates the constraints for the peak fields, corresponding to the planes where the HTS peak field reaches 18.5 T and the LTS peak field reaches 12.4 T.

The optimization result is as follows: The number of turns of HTS1U is 5×48 , and the number of turns of HTS2U and HTS3U is 5×23 . For LTS1U, the number of turns is 11×48 , while the number of turns for LTS2U and LTS3U is 11×23 . With an operation current of 39.5 kA, the peak fields on HTS1U, HTS2U, and HTS3U are 18.5, 17.6, and 16.2 T, respectively. Meanwhile, the peak fields of LTS1U, LTS2U, and LTS3U are 12.2, 11.6, and 10.6 T, respectively. The magnetic flux is 54.6 V s, while the WP height of the CS coils is 7927.6 mm. The total conductor lengths are 3745 m and 11 196 m for the HTS and LTS sub-coils, respectively. The key parameters of the final optimization design are listed in Table II.

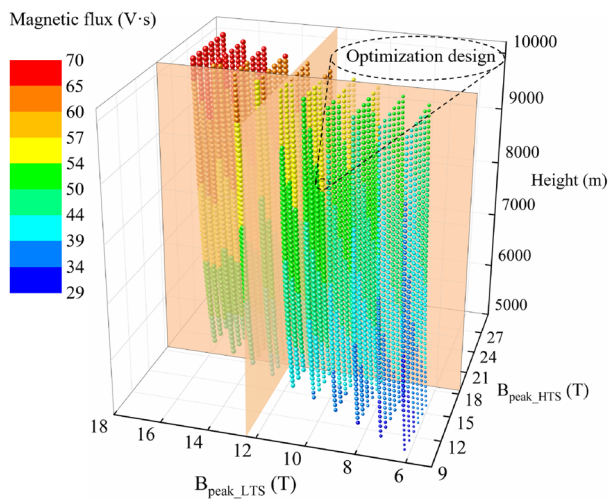


FIG. 5. 3D map of electromagnetic design optimization of the CS coil.

TABLE II. Key parameters of the optimization design.

	HTS module	LTS module
Number of turns of 1U	5×48	11×48
Number of turns of 2U/3U	5×23	11×23
Outer radius (mm)	698.4	1000.0
Inner radius (mm)	478.4	723.4
Iop (kA)	39.5	
Peak field on 1U (T)	18.5	12.2
Peak field on 2U (T)	17.6	11.6
Peak field on 3U (T)	16.2	10.6
Conductor length (m)	3745	11 196
Height of WP (mm)	7927.6	
Magnetic flux (V s)	54.6	

To verify the result of the optimization design obtained with our self-written code, the magnetic field and magnetic flux are calculated and assessed by using Maxwell electromagnetic analysis.

IV. ELECTROMAGNETIC ANALYSIS

As shown in Fig. 6, the Maxwell electromagnetic analysis shows that the peak field of the HTS (LTS) sub-coil is located on HTS1U (LTS1U), which is 18.3 T (12.3 T). As listed in Table III, the peak fields and the magnetic flux obtained using Maxwell are in good agreement with the results of our code.

The fast discharge of the CS coil is triggered by several dump resistors, as shown in Table IV. Due to the strong magnetic coupling between the HTS and LTS sub-coils,³⁵ the mutual inductance

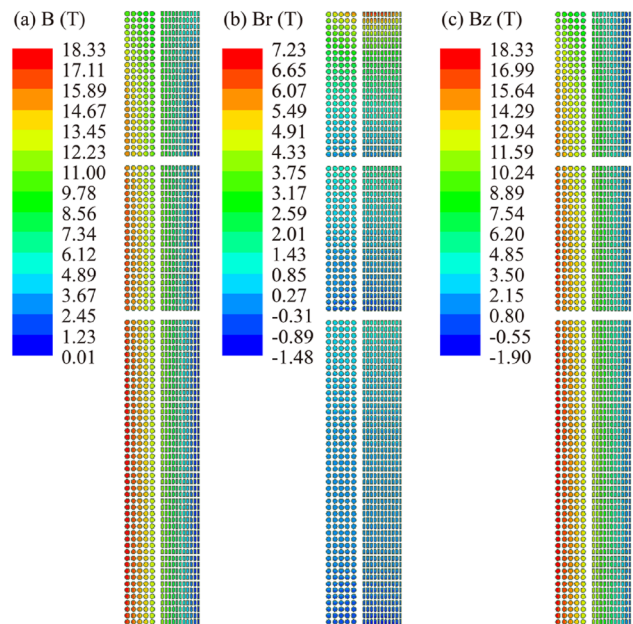


FIG. 6. Verification of the peak fields with Maxwell. (a) Magnetic field B, (b) radial magnetic field Br, and (c) axial magnetic field Bz.

TABLE III. Verification of optimization result by using ANSYS.

	Self-written code	Maxwell	Diff. (%)
B _{HTS1U} (T)	18.5	18.3	1.1
B _{HTS2U} (T)	17.6	17.5	0.6
B _{HTS3U} (T)	16.2	16.1	0.6
B _{LTS1U} (T)	12.2	12.3	0.8
B _{LTS2U} (T)	11.6	11.7	0.9
B _{LTS3U} (T)	10.6	10.6	0.0
Magnetic flux (V s)	54.6	54.6	0.0

TABLE IV. Dump resistors of HTS and LTS sub-coils.

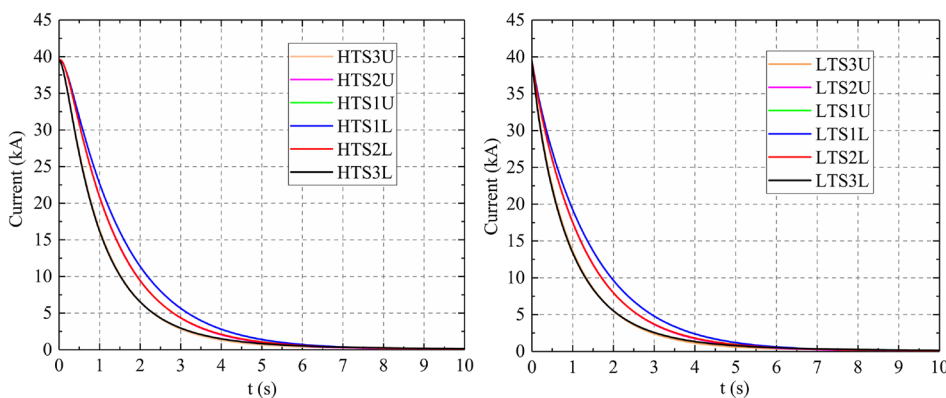
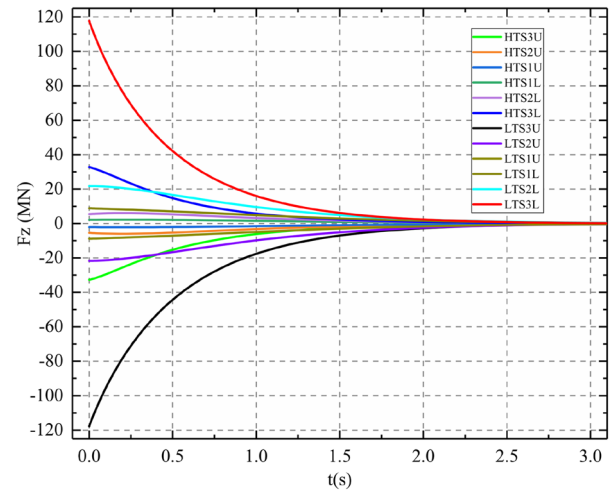
	Resistor (Ω)
HTS 1U/1L	0.070
HTS 2U/3U/2L/3L	0.035
LTS 1U/1L	0.300
LTS 2U/3U/2L/3L	0.150

between the sub-coils should be considered when calculating the current during fast discharge. The circuit equation can be expressed as

$$L_i \frac{dI_i(t)}{dt} + \sum_{j=1, j \neq i}^{12} M_{ij} \frac{dI_j(t)}{dt} + R_i \cdot I_i(t) = 0, \quad (1)$$

where L_i is the self-inductance, M_{ij} is the mutual inductance between CS coils, I_i and I_j are the current of sub-coils during fast discharge, and R_i is the dump resistor resistance.

As shown in Fig. 7, the operating current undergoes exponential decay to zero during fast discharge, and the rates of decrease for the HTS and LTS sub-coils differ due to their different inductances. When the coil current drops, the induced current will be generated on other coils, and the coil with less self-induction will be affected more, which will cause the affected coil current to increase slightly and then drop slowly, such as HTS1L and HTS1U.

**FIG. 7.** Operation current during fast discharge of HTS and LTS sub-coils.**FIG. 8.** Vertical force on HTS and LTS sub-coils during fast discharge.

The Lorentz force on the CS sub-coils during fast discharge can be expressed as

$$F_i = \sum_{j=1, j \neq i}^{12} f_{ij} \cdot I_i \cdot I_j, \quad (2)$$

where F_i is the vertical Lorentz force and f_{ij} is the force coefficient.

As illustrated in Fig. 8, the vertical forces on the sub-coils during fast discharge decrease to zero rapidly within 3 s.

V. MECHANICAL ANALYSIS OF CONDUCTOR JACKET

To further validate the jacket hoop stress calculated using our code, mechanical analysis is performed with ANSYS.³⁶ A two-dimensional finite-element model is built based on the final optimization design, as shown in Fig. 9. In addition, the material properties are shown in Table V.

The contact between the conductor and insulation is set as bonded, which should be a reasonable assumption because after vacuum-pressure impregnation (VPI) of the winding pack, the conductor and the insulation will be bonded. If these contacts are set as

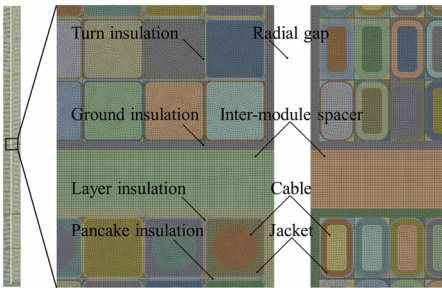


FIG. 9. Finite element model of the final optimization design.

frictional, the finite element mechanical analysis would become challenging due to the excessive nonlinearity of the contacts. The contacts between the module ground insulation and the inter-module spacer is set to be frictional, with a friction coefficient of 0.25.⁶ The displacement of the bottom support is fixed in the axial direction and free in the radial direction. The mechanical analysis is performed for the 39.5 kA energizing condition considering the preloads applied to HTS and LTS, which are 0.1 and 0.2 MPa, respectively (the dead load and the cooldown load are also applied).

TABLE V. Material properties used in the 2D finite element model of the CS coil.

Properties	Jacket	Cable	Insulation	Resin filler
E_x (GPa)	205	1	12	15
E_y (GPa)	205	1	20	15
E_z (GPa)	205	1	20	15
ν_{xy}	78.8	0.38	6	6
ν_{yz}	78.8	0.38	6	6
ν_{xz}	78.8	0.38	6	6
G_{xy} (GPa)	0.3	0.3	0.33	0.3
G_{yz} (GPa)	0.3	0.3	0.17	0.3
G_{xz} (GPa)	0.3	0.3	0.33	0.3
CTE_X	0.29%	0.29%	0.7%	0.5%
CTE_Y	0.29%	0.29%	0.25%	0.5%
CTE_Z	0.29%	0.29%	0.25%	0.5%

The peak stress intensities on the conductor jackets under various load conditions are displayed in Fig. 10. At the energizing condition of $t = 0$ s, the stress intensities of HTS and LTS jackets are 923.9 and 1252.2 MPa, respectively. For the HTS conductor jacket,

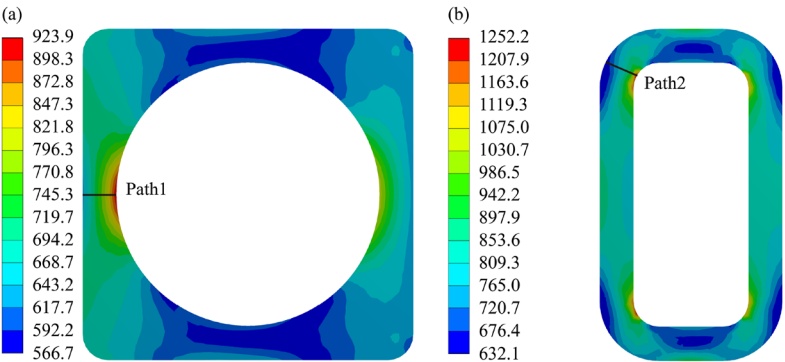


FIG. 10. Stress intensity (MPa) distribution at energizing $t = 0$ s of (a) HTS jacket and (b) LTS jacket, generated by 2D finite element analysis.

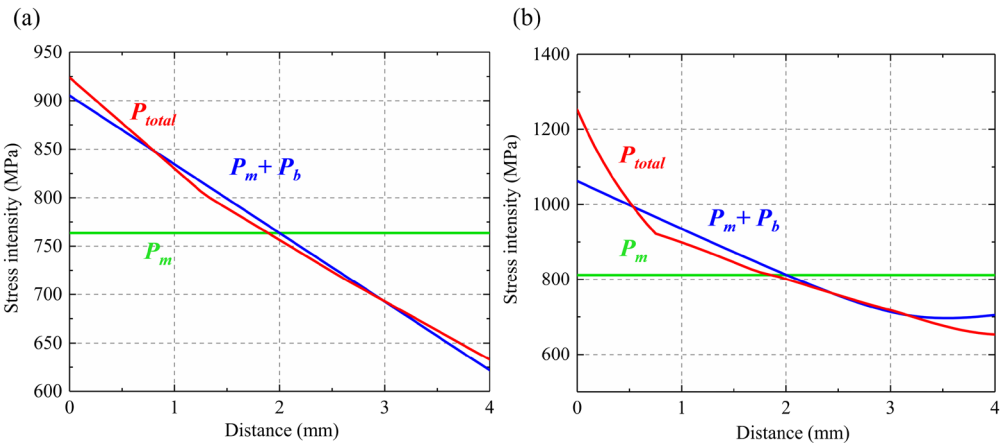


FIG. 11. Stress linearization of (a) HTS jacket and (b) LTS jacket at energizing $t = 0$ s.

P_m is 763.5 MPa and $P_m + P_b$ is 905.3 MPa. For the LTS conductor jacket, P_m is 812.1 MPa and $P_m + P_b$ is 1062.3 MPa, as shown in Fig. 11. According to the ITER magnet structural design requirements,³⁷ these are below the allowable limits, where the allowable P_m is 933.3 MPa and the allowable $P_m + P_b$ is 1213.3 MPa, considering the 1400 MPa yield strength of N50H at 4 K.

VI. CONCLUSIONS

The design of a hybrid CS coil is optimized to satisfy the requirement of magnetic flux for the next generation fusion device. The peak field, magnetic flux, and total height of the CS winding pack are taken into account. The magnetic flux of the final optimization design is 54.6 V s. In addition, the results are verified by using Maxwell. The difference between the two results is smaller than 1.1%, which proves the accuracy of the self-written MATLAB code. Then, the operating current and vertical force on the sub-coils during fast discharge are calculated. Subsequently, the finite-element mechanical analysis is conducted, confirming compliance with the ITER magnet structural design requirements. In the next steps, more detailed tasks will be undertaken, including the preload structural design and analysis, as well as static and fatigue mechanical analyses of the conductor jackets.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Jiandong Zhu: Data curation (equal); Formal analysis (equal); Writing – original draft (equal). **Xiaogang Liu:** Writing – review & editing (equal). **Xiang Gao:** Resources (equal). **Jie Zhang:** Resources (equal). **Lijuan Yu:** Data curation (equal). **Fanping Ding:** Investigation (equal). **Guoqiang Li:** Project administration (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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